

Patalee Narasinghe

AI & Cloud Engineering Enthusiast



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🐙 GitHub | 🌐 Portfolio

Data Science Engineering undergraduate with hands-on experience in **AWS Cloud architecture**, **MLOps pipelines**, **Generative AI**, and **Vision**. Proven track record of Architecting cloud infrastructure, automating deployments using Terraform and CI/CD, and fine-tuning Foundation Models for production environments.

Education

University of Moratuwa

B.Sc. (Hons) in Engineering, Computer Science & Engineering

Specialization: Data Science Engineering | Minor in Mathematics

CGPA: 3.66/4.00

Aug 2022 - Present

Expected Graduation: 2026

Professional Experience

Software Engineering Intern (Cloud & AI) | 1 year +

Dec 2024 – Present

Pekoe.ai

- 🕒 **MLOps & CI/CD:** Designed an automated ML deployment pipeline on **AWS** (Step Functions, Lambda) using **Terraform** for IaC, ensuring consistent delivery of quality analysis models.
- 🕒 **Data Engineering:** Architected a serverless system handling 12,000+ weekly images, Created datasets for model finetuning.
- 🕒 **Agentic AI Systems:** Developed a multi-agent solution with **AWS Bedrock** that automates report generation, orchestrating LLMs to synthesize analysis data into actionable insights.
- 🕒 **Full Stack Development:** Built **React** dashboards integrated with serverless Python backends
- 🕒 **Computer Vision:** Applied Transfer Learning to vision models for feature extraction, fine-tuning performance on limited tea leaf image datasets.

Technical Skills

Cloud & DevOps

AWS (Lambda, EC2, S3, Step Functions, CodePipeline), Docker, Terraform, Ansible

AI & ML

PyTorch, Spark, Scikit-learn, LLMs, PEFT/LoRA, MLOps, Explainable AI (XAI)

Languages

Python, JavaScript/TypeScript, Java, C

Web & Databases

React, Node.js, Flask, FastAPI, MongoDB, MySQL, PostgreSQL

Key Technical Projects

Explainable Voice Analysis Framework for Speech Models | Final year project

2025-Present

- 🕒 Designing a model-agnostic explainability framework for speech models (WavLM, Wav2Vec) with layer-wise and neuron-wise analysis.
- 🕒 Integrating **probing techniques**, SHAP, and attention head analysis with **Sparse Autoencoders (SAEs)** to disentangle neurons for precise **neuron-level causal analysis**.
- 🕒 Developing an interactive visualization module using Flask and JavaScript for model interpretability and comparison.
- 🕒 **Tech Stack:** Python, PyTorch, Flask, JavaScript.

Few-Shot Adaptation of Multimodal Foundation Models | Accepted at **VISAPP 26**

2025

- Paper accepted at CORE **rank B conference** , VISAPP 2026
- Conducted a comprehensive empirical study on adapting Contrastive Captioners for low-resource visual recognition, uncovering insights.
- Implemented and evaluated PEFT methods including **LoRA**, hybrid prototype classification, and linear probing.
- Achieved optimal results by combining **Supervised Contrastive with cross-entropy**.
- **Tech Stack:** PyTorch, Transformers, LoRA, Metric Learning.

Sentiment Analysis & Product Improvement Engine (NLP)

2024

- Developed a full-stack NLP solution to extract pain points from Amazon reviews using **BERT** for classification and **LSTM** for emotion detection.
- Integrated **LLAMA 3** to generate actionable business recommendations and **LIME** for model interpretability.
- **Tech Stack:** Python, Hugging Face, AWS EC2,S3, Flask, React.js, CI/CD.

Agentic AI Report Generation Engine

2024

- Designed a serverless **Orchestrator Lambda** to parse JSON report schemas and dynamically dispatch content-level analysis tasks to specialized AI agents.
- Integrated **AWS Bedrock** to power agents that autonomously query **Knowledge Bases** and SQL databases to retrieve context-aware domain data.
- **Tech Stack:** AWS Bedrock, Lambda, AppSync, S3, Python, SQL, React.js

Real-Time Voice Translator & Converter

2022

- Built a low-latency voice processing app using **AsyncIO**, integrating OpenAI and Deepgram APIs for speech-to-text and voice conversion.
- **Tech Stack:** Python, PyQt5, PyAudio, Asynchronous API Integration.

Certifications

- ✳ **Aviatrix Multi-cloud Network Associate** (2025)
- ✳ **Nvidia Fundamentals of Deep Learning** (2025)
- ✳ **KodeKloud Terraform for Beginners** (2025)
- ✳ **AWS Academy ML Foundations** (2024)
- ✳ **AWS Intro to Generative AI** (2025)
- ✳ **Kaggle Intro to Machine Learning** (2024)
- ✳ **Kaggle Intermediate Machine Learning** (2024)
- ✳ **Udemy Intermediate Python Training** (2021)

Achievements

- 🏆 **Runners-up** - IESL Techno Hackathon 2023
- 🏆 **Finalist** - Cypher Hackathon 2023 & 2024
- 🏆 **6th Place** - SLIIT Bashway 2023
- 🏆 **Finalist** - SLIIT Codefest 2018

References

Prof. Uthayasanker Thayasivam
Supervisor - Final year Research
Dept. of CSE
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